1.3: Essential Active Components

Introduction

Welcome to this module on **Essential Active Components**! In the world of electronics, components are broadly categorized into two types: passive and active. Passive components, like resistors, capacitors, and inductors, do not require an external power source to operate and cannot amplify or generate power. Active components, on the other hand, require an external power source and are capable of controlling, amplifying, or generating electrical signals. They are the 'brains' of electronic circuits, enabling complex functionalities.

This module will focus on two fundamental active components: **diodes** and **transistors**. We will explore their basic principles, how they function in circuits, and their primary applications, such as rectification using diodes and switching/amplification using transistors.

1. Diodes

A **diode** is a two-terminal electronic component that primarily conducts current in one direction. It acts like a one-way valve for electricity. The two terminals are called the **anode** (positive) and the **cathode** (negative).

1.1. Diode Structure and Symbol

A diode is typically made from semiconductor materials, most commonly silicon. It consists of a p-n junction, where a p-type semiconductor (rich in holes) is joined with an n-type semiconductor (rich in electrons).

Circuit Symbol:

```
Anode (+) ----|>|---- Cathode (-)
```

The arrow in the symbol indicates the direction of conventional current flow when the diode is forward-biased.

1.2. Diode Operation (Forward and Reverse Bias)

Forward Bias

When the anode is connected to a higher positive voltage than the cathode, the diode is **forward-biased**. This reduces the depletion region at the p-n junction, allowing current to flow easily through the diode once a certain voltage (known as the **forward voltage drop** or **cut-in voltage**, typically around 0.7V for silicon diodes) is exceeded.

Reverse Bias

When the cathode is connected to a higher positive voltage than the anode, the diode is **reverse-biased**. This widens the depletion region, effectively blocking current flow. Only a very small leakage current flows in reverse bias, which is usually negligible. If the reverse voltage becomes too high, the diode can break down and conduct current in the reverse direction, which can damage the diode (unless it's a Zener diode designed for this purpose).

1.3. Rectification

One of the most common applications of diodes is **rectification**, the process of converting alternating current (AC) to direct current (DC). AC voltage periodically reverses direction, while DC voltage flows in only one direction. Diodes achieve rectification by allowing current to flow during one half-cycle of the AC input and blocking it during the other half-cycle.

Half-Wave Rectifier

A half-wave rectifier uses a single diode to convert AC to pulsating DC. It allows only one half-cycle of the AC waveform to pass through.

Input AC Waveform:

```
V_{in}(t) = V_{peak} \sin(\omega t)
```

Output DC Waveform (ideal diode):

```
V_{out}(t) = \begin{cases}
V_{in}(t) & \text{if } V_{in}(t) > 0 \\
0 & \text{if } V_{in}(t) \le 0
\end{cases}
```

Full-Wave Rectifier

A full-wave rectifier is more efficient as it converts both half-cycles of the AC input into pulsating DC. This can be achieved using a center-tapped transformer with two diodes or a bridge rectifier with four diodes.

Comparison of Rectifiers:

Feature	Half-Wave Rectifier	Full-Wave Rectifier (Bridge)
Number of Diodes	1	4
Output Efficiency	Lower	Higher
Ripple Frequency	Equal to input freq	Twice the input freq
Transformer	Optional	Optional

1.4. Practice Problem (Diodes)

Problem: A silicon diode has a forward voltage drop of 0.7V. If it is connected in series with a 100 Ω resistor and a 5V DC supply, calculate the current flowing through the circuit when the diode is forward-biased.

Solution:

Given:

- Supply Voltage (V_S) = 5V
- Diode Forward Voltage Drop (V_D) = 0.7V
- Resistor (R) = 100 Ω

Using Kirchhoff's Voltage Law (KVL) for a series circuit:

```
V_S = V_D + I \cdot R
```

Rearranging to solve for current (I):

```
I = \frac{V_S - V_D}{R}
```

Substitute the values:

```
I = \frac{5V - 0.7V}{100 \Omega} = \frac{4.3V}{100 \Omega} = 0.043 A
```

Therefore, the current flowing through the circuit is 0.043 Amperes or 43 mA.

2. Transistors

A **transistor** is a semiconductor device used to amplify or switch electronic signals and electrical power. It is one of the fundamental building blocks of modern electronic devices. Transistors come in various types, but we will focus on Bipolar Junction Transistors (BJTs) and Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs).

2.1. Bipolar Junction Transistors (BJTs)

BJTs are current-controlled devices. They have three terminals: the **base** (B), **collector** ©, and **emitter** (E). There are two main types: NPN and PNP.

NPN Transistor Symbol:

```
      Collector (C)
      |
      |/
      |---- Base (B)
      |\
      |
      Emitter (E)
```

PNP Transistor Symbol:

```
Collector (C)
|
|\
|---- Base (B)
|/
|
Emitter (E)
```

BJT Operation (as a Switch)

In switching applications, a BJT operates in two main regions: **cutoff** and **saturation**.

- **Cutoff Region:** When no current or very little current flows into the base, the transistor acts like an open switch, and no current flows from collector to emitter.
- **Saturation Region:** When sufficient current flows into the base, the transistor acts like a closed switch, allowing maximum current to flow from collector to emitter.

BJT Operation (as an Amplifier)

In amplification applications, a BJT operates in the **active region**. A small change in base current causes a much larger change in collector current, thus amplifying the input signal.

2.2. Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs)

MOSFETs are voltage-controlled devices. They also have three terminals: the **gate** (G), **drain** (D), and **source** (S). There are two main types: n-channel and p-channel, and each can be further classified as enhancement-mode or depletion-mode.

n-Channel Enhancement-Mode MOSFET Symbol:

```
Drain (D)
|
|\
|---- Gate (G)
|/
|
Source (S)
```

MOSFET Operation (as a Switch)

Similar to BJTs, MOSFETs can act as switches:

- **Cutoff Region:** When the gate-source voltage (V_{GS}) is below the threshold voltage (V_{th}), the MOSFET is off, acting as an open switch.
- **Triode/Ohmic Region (for switching):** When V_{GS} is sufficiently above V_{th}, the MOSFET is fully on, acting as a closed switch with low resistance.

MOSFET Operation (as an Amplifier)

For amplification, MOSFETs operate in the **saturation region** (different from BJT saturation). In this region, a small change in V_{GS} results in a controlled change in drain current (I_D), providing amplification.

2.3. Comparison of BJTs and MOSFETs

Feature	BJT (Bipolar Junction Transistor)	MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor)
Control Mechanism	Current-controlled (Base current)	Voltage-controlled (Gate-Source voltage)
Input Impedance	Low	Very High
Switching Speed	Generally slower	Generally faster
Power Dissipation	Higher (due to base current)	Lower (due to high input impedance)
Applications	Amplifiers, switches	High-speed switches, power amplifiers, digital circuits

2.4. Practice Problem (Transistors)

Problem: An n-channel enhancement-mode MOSFET has a threshold voltage (V_{th}) of 2V. If it is used as a switch and the gate-source voltage (V_{GS}) is 0V, what is the state of the MOSFET (on or off)? What if V_{GS} is 5V?

Solution:

Given:

- Threshold Voltage (V_{th}) = 2V

Case 1: V_{GS} = 0V

Since V_{GS} (0V) is less than V_{th} (2V), the MOSFET is in the **cutoff region**. Therefore, the MOSFET is **off** and acts as an open switch.

Case 2: V_{GS} = 5V

Since V_{GS} (5V) is greater than V_{th} (2V), the MOSFET is in the **triode/ohmic region** (or saturation if used as an amplifier, but for switching, it's considered fully on). Therefore, the MOSFET is **on** and acts as a closed switch.

Conclusion

Diodes and transistors are indispensable active components in electronics. Diodes, with their unidirectional current flow property, are crucial for rectification, converting AC to DC. Transistors, particularly BJTs and MOSFETs, serve as versatile switches and amplifiers, forming the backbone of almost all modern electronic devices, from simple logic gates to complex microprocessors. Understanding their fundamental operation is key to designing and analyzing electronic circuits.

References

[1] Robert L. Boylestad, Louis Nashelsky. *Electronic Devices and Circuit Theory*. Pearson Education. [2] Adel S. Sedra, Kenneth C. Smith. *Microelectronic Circuits*. Oxford University Press. [3] Donald A. Neamen. *Semiconductor Physics and Devices*. McGraw-Hill Education.